Xingjian Qi

Institute of Computer Engineering (ZITI), Heidelberg University
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EDUCATION

Heidelberg University Jan 2024–Present
PhD in Computer Engineering *Advisor: Robert Strzodka*

Imperial College London Oct 2022–Oct 2023
MSc in Applied Computational Science and Engineering

University of Illinois Urbana–Champaign Sep 2018–May 2020
MS in Civil and Environmental Engineering (Transportation)

Harbin Institute of Technology Sep 2015–Jun 2019
BS in Transportation Engineering

PUBLICATIONS

[1] **Xingjian Qi** and Robert Strzodka. "VILIB++: High-Performance C++ Library for Auto-Vectorizing Interval Arithmetic." *55th International Conference on Parallel Processing (ICPP '26)*.

[2] **Xingjian Qi** and Robert Strzodka. "Evaluating Function-Level Accuracy and Performance of SIMD Math Libraries with FP32 Inputs." *CORRECTNESS: 10th International Workshop on Software Correctness for HPC Applications Session of the SC Workshops '26*, accepted, 2026.

RESEARCH EXPERIENCE

VILIB++: Auto-Vectorizing Interval Arithmetic

- Designed a branchless outward-rounding kernel that avoids global rounding-mode changes and enables compiler vectorization in a drop-in C++ interval type.
- Demonstrated up to 12× arithmetic speedup over existing interval libraries and hide interval overhead in memory-bound reductions, small 1D stencils, and Horner polynomials up to degree 12.
- Our method introduces less than 2× error for basic binary arithmetic operations.

Exhaustive FP32 SIMD Math-Library Evaluation

- Built an accuracy and performance framework that exhaustively tests all 4,278,190,080 finite FP32 encodings per function–library pair against a 256-bit MPFR oracle across six AOCL, libmvec, SLEEF, and SVML implementations.
- Identified documented accuracy-bound violations and showed that data-dependent paths can change throughput by an order of magnitude.

VLIES/VLIESFP: Tapered Floating-Point Formats *Ongoing*

- Designed a configurable variable-length integer encoding and a tapered floating-point framework.
- Implemented the representation and arithmetic operations in a C++ emulation library with tunable precision–dynamic-range trade-offs.

GPU Implementation of Mixed Precisiton Multilevel Monte Carlo *Ongoing*

- Developing a GPU implementation of a mixed precision multilevel Monte Carlo algorithm.

Shared-Memory Parallel Patterns *for Tutorials*

- Built TBB and OpenMP reference implementations of common parallel patterns, including NUMA-aware array-of-structures and structure-of-arrays transformations.

PROFESSIONAL EXPERIENCE

Algorithm Scientist Aug 2021–Aug 2022
Yum China Holdings, Inc.

- Developed several mixed-integer linear programming model to optimize and support daily operation of the company.
- For example, developed a mixed-integer linear programming model for distributing chicken from suppliers to warehouses, balancing days of service while satisfying supply, forecast, delivery-cost, and operational constraints. This reduces operating costs by CNY 200,000 (6% of total cost) per month compared with the previous method.

TEACHING EXPERIENCE

CPU Algorithm Design — Tutorials
Heidelberg University

SERVICE

I'm part of the IEEE P754 working group.

TECHNICAL SKILLS

Languages and Tools	C++, C, Python, CUDA, OpenMP, MPI, TBB, xsimd
Packages	MPFR , networkx, PyTorch, NumPy, pandas, gurobipy, cupy, LaTeX